

2898 A.D: THE SERUM HUNT

STORY-

2898 A.D. The world is ash. Supreme Yaskin, ruler of the Complex, has learned from his Counsellor Bani—a mythologist who decoded ancient Dwarka texts—that fossils from the sunken city hold the key to immortality. The Complex Army (Raiders) found those fossils and they were in return, Rebels from Shambhala attacked to prevent the fossils from Yaskin such that world will be saved but with advanced weapons like plasma blasts and tech, Raiders won the Bombay War in 2896 A.D and Rebels were injured critically.

But in 2898, a Rebel spy within the Complex stole the completed serum and destroyed the lab data. The serum now lies hidden in Shambhala—a city no outsider has found. Commander Manus has announced a bounty of 1 Million Units and Complex citizenship to any hunter who retrieves it.

You are a bounty hunter. Your route: Kashi → Sector 7 → New World → Shambhala. The Rebels have laid traps. Fall, and you vanish into the Flux Lands where Nomen feed. Succeed, and Ashwatthama himself—the immortal warrior of Mahabharata—must let you pass.

Here Volunteers are Raiders, players are bounty hunters.

Objectives

- Present Electronics concepts in a crowd-friendly and story-driven format.
- Attract both technical and non-technical students to the Electronics department stall.
- Deliver a high-value experience relative to the entry fee.
- Ensure the game is fast enough for continuous participation during a fest.
- Use reusable props and low-reset mechanics suitable for hall-based operation.

This event consists of 4 levels, each level having its own ECE subjects-based games.

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EXPLANATION:

Level 1: Kashi - SOMA Activation

The Lore: *You are in Kashi preparing for the journey. You need to activate the ancient SOMA energy drink dispenser to boost your life support, but the grid is fluctuating.*

The Game (The Circuit Maze): * Have a large, beautifully painted wooden board or a digital screen showing a grid of resistors/switches.

To get the SOMA drink, the player must find the **"Path of Least Resistance"** (Short Circuit) from Point A to Point B.

How it works: Instead of complex calculations, it's a **physical wire-loop game** or a **switch-flipping puzzle**.

There are 3 possible paths. Path 1 has 3 glowing bulbs (High resistance = low energy). Path 2 has 2 bulbs. Path 3 has a direct copper wire (Zero resistance). The player just has to physically connect the jumper wire to the path with *zero bulbs* to trigger a green light and "drink" the SOMA.

Level 2: Sector 7 - Raider Logic Grid

The Lore: *You've reached Sector 7, a heavy Complex checkpoint. To bypass the Raider barricade, you must hack their digital gate using a binary code combination before the automated plasma turrets lock onto you.*

The Game (The Binary Gatekeeper):

You have a control panel with **4 giant toggle switches** (representing bits: 8, 4, 2, 1). Above the switches, a digital display (or a painted board) shows a decimal number—let's say the bounty amount: **"Target Code: 11"** (or 13, 9, etc., randomized for every player using a simple Arduino setup or just cards you flip).

How it works for General Audience:

Next to the switches, have a cool, thematic cheat-sheet styled like a "Raider Hacking Manual".

Manual text: Switch 1 = 8 points | Switch 2 = 4 points | Switch 3 = 2 points | Switch 4 = 1 point.

To get "11", the player needs to flick Switch 1 (8), Switch 3 (2), and Switch 4 (1) to the "ON" position ($8+2+1=11$). When the correct combination is flipped, a magnetic lock opens the door to the next zone. It's simple addition for them, but they just practiced **Binary to Decimal conversion!**

Time Limit: 60 seconds

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Level 3: New World - Rebel Signal Intercept

The Lore: Entering the communication-heavy New World, the Rebels have broadcasted a heavily distorted cloaking frequency to hide Shambhala's coordinates. You must clear the noise to lock onto their transmission.

ECE Concept: Signals & Systems (Waveform Matching & Signal-to-Noise Ratio).

Gameplay: A monitor or tablet displays a clear target "Rebel Waveform" (a clean sine wave). The player controls a physical dial (potentiometer connected to an Arduino or a software slider). Turning the dial modifies a secondary, chaotic waveform on screen.

How it Works: It is a sensory, visual puzzle. The player turns the knob until their messy signal cleans up and perfectly overlaps with the static target waveform. Once aligned, a tracking lock sound effect plays.

Level 4: Shambhala Gate - Real Serum Verification

Two-Phase Finale:

Phase 1 (90 sec): The Binary Mantra

- Ashwatthama speaks a "mantra" (recorded voice): *"I am from the age of 18 days, 18 armies, 18 chapters"*
- Player must convert **18** to binary (10010) using 5 switches
- Then calculate **2's complement** of that binary (01110) using provided formula card
- Enter both on dual display

Phase 2 (90 sec): The Network of Eternity

- Physical board with **resistor network** (5 resistors, 2 voltage sources)
- Player given: "The serum requires exactly 5V to activate. The source provides 9V. Design the voltage divider."
- Must place **2 resistors** in series from provided options (1k Ω , 2k Ω , 4k Ω , 10k Ω)

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- Formula: $V_{out} = V_{in} \times (R_2 / (R_1 + R_2))$
- Correct answer: $R_1 = 4k\Omega$, $R_2 = 1k\Omega \rightarrow 9V \times (1/5) = 1.8V$... wait, that's wrong.

Correct math for 5V from 9V:

- Need ratio: $5/9 = R_2 / (R_1 + R_2)$
- $R_2 = 5k\Omega$, $R_1 = 4k\Omega \rightarrow 9 \times (5/9) = 5V$ ✓
- So provide: 1k, 2k, 4k, 5k, 10k options

Twist: Board has a "Flux Land" trap—one resistor pair creates a **short** (0V). Player must use multimeter prop to test before final connection.

Fail condition: 2 wrong attempts → "Nomen feed" (Game Over)

FAILURE STATE: THE FLUX LANDS- Any level failure diverts the hunter to the Flux Lands—a dark zone where Nomen (volunteers in masks) claim their due. One retry is granted by sacrificing all remaining SOMA energy.

Format: Solo gamified technical experience stall

Theme: Dystopian sci-fi mission inspired by the world-building of *Kalki 2898 AD*

Target audience: All departments, not limited to Electronics students

Recommended duration per player: 6 to 8 minutes

We need to setup props for the dystopian world setting.

Prizes: Laptop stickers of KALKI 2898 A.D or Acrylic keychain of KALKI 2898 A.D

Components Required:

Item	Quantity	Purpose
4 toggle switches or push buttons	4	Input binary (0/1)
5 LED indicators	5	Show binary input (ON/OFF)

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Item	Quantity	Purpose
3-digit keypad or 10 push buttons (0–9)	1	Enter decimal code
9V battery + battery clip	1	Power circuit
560Ω resistors	5	LED current limiting
Laminated binary hint card	1	Show pattern to convert
Foam board or cardboard panel	1	Mount all components
Buzzer	1	Unlock sound
Small lock or magnetic latch	1	Physical unlock for scanner
Laptop or tablet	1	Display waveforms
Printed waveform cards (clean vs noisy)	4–6	Physical alternative to screen
Speaker or small audio buzzer	1	Play pulse sequences
Toggle switches or arrow buttons	2–3	Select signal option
LED green + red	2	Correct/incorrect feedback
Foam board panel	1	Mount switches and LEDs
9V battery + clip	1	Power
Toggle switches (SPDT)	3–4	Input A, B, C
LEDs (green)	2–3	Show route open/closed

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Item	Quantity	Purpose
9V battery + clip	1	Power
Resistors 560Ω	3	LED limiting
Clue card with logic condition	1	"Open Route A if A AND B are ON"
Printed board with nodes and weighted edges	1	Path maze
Wire with connectors or path tokens	6–8	Trace route
Battery 9V + clip	1	Power circuit
LED green	1	"Valid path" feedback
Small DC power module or 555 timer	1	Circuit completion detector
foam board base	1	Mount board

Props:

Item	Quantity	Purpose
Black cloth or dark paper	Based on requirement	Backdrop for dystopian feel
Red neon LED strip	2	Mood lighting
Sound speaker + Bluetooth	1	Alarm, radio chatter, bass drones
Plastic/glass vials (4 total)	4	Serum props (1 real, 3 fake)
Glow paint or LED inside vial	1	Real serum glows
Bounty pass cards (printed)	50+	Entry pass for players
Raider stamp or sticker	Based on requirement	Level completion mark
Envelopes with game/movie stamp	50+	Reward envelope
Stickers (winners)*	30+	Keychain-style or badge stickers
Stickers (non-winners)*	20+	Single sticker for partial finish
Keychains (winners)*	15–20	Premium reward
Poster prints ("NOW BEGINS THE END", "PROJECT K")	4–6	Queue marketing

* either stickers or keychains not both, if keychains then non-winners will get envelop with a colour poster.